Nonvanishing of the common 2+2 source function

Formal proof packet

28 October 2026

This packet reports conditional Lean declarations on the actual weighted-Proj curve. It contains no new DLP trial or cost ratio.

Statement and mechanism

Let K be a field and let the affine chart be $K[x, y]/(y^2 - f(x))$. The checked nested polynomial equivalence maps the defining relation to a polynomial of degree two in y. An x-only polynomial has degree zero in y. The new affine lemmas prove $y \neq p(x)$ and $p(x) - By \neq 0$ for $B \neq 0$.

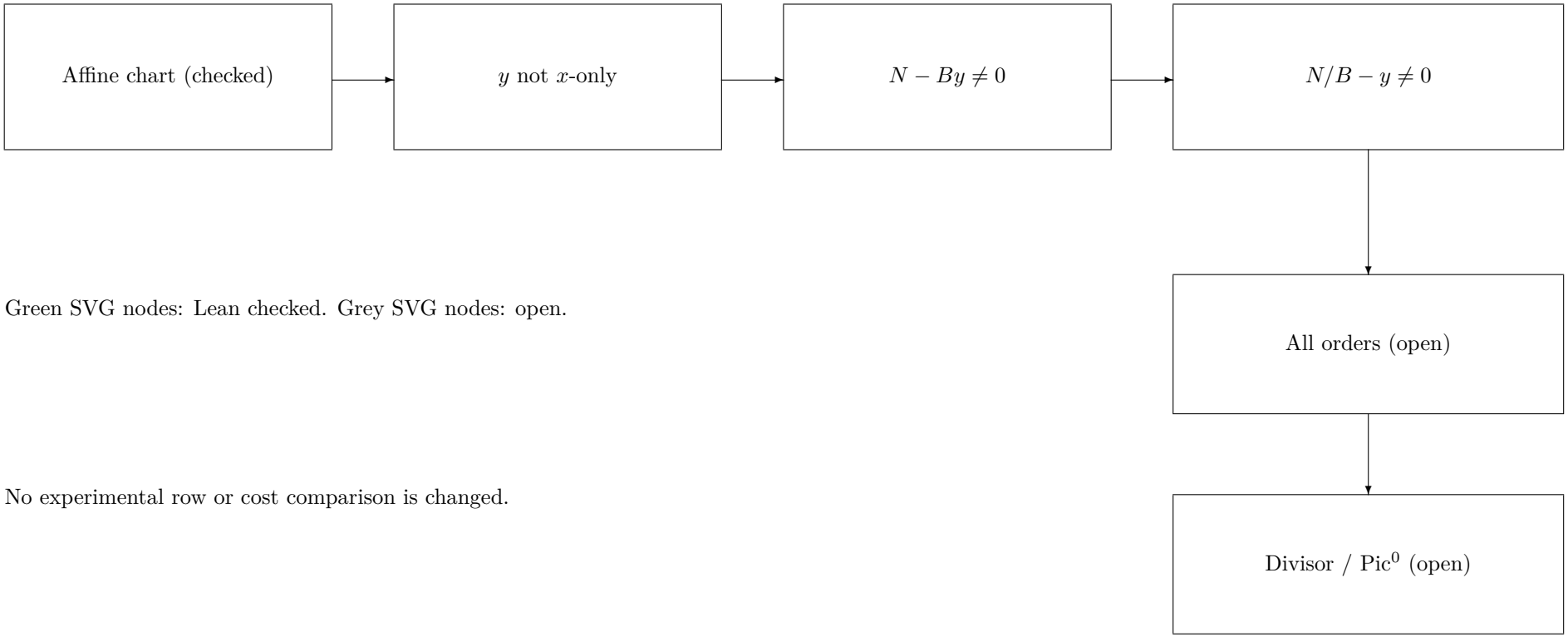
The common 2+2 numerator N is x-only. A checked specialization gives $N - By \neq 0$. The existing injection of this chart into the function field of the actual weighted-Proj curve gives $N/B - y \neq 0$ when the quintic is separable and $B \neq 0$. The affine degree argument needs no separability or characteristic restriction.

Validation and limits

Pinned Lean 4.34.1 checked all four declarations. Direct axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`. The standardized CSV separates checked statements from open gates. These include closed-point coheight, every finite and infinity order, a principal divisor, the Pic^0 class equation, and compiled Rust correspondence. The prior 1,260-target cohort and matched-rho cost rows are unchanged.

Prior source: the existing chart-degree proof in `Common22BranchDomain.lean`; the existing embedding in `WeightedProjSourceFunctionField.lean`; Stacks Project Lemma 10.119.7 for the downstream DVR valuation framework (not used to assert the missing global divisor).

Dependency diagram



Green SVG nodes: Lean checked. Grey SVG nodes: open.

No experimental row or cost comparison is changed.